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dython: A Set of Analysis and Visualization Tools for Data and Variables in Python

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Software

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Summary

Exploratory data analysis (EDA) frequently requires quantifying and visualizing associations between variables. In datasets with mixed variable types (continuous and categorical), analysts must manually choose suitable metrics (Pearson's R, correlation ratio, Cramér's V, Theil's U, etc.), compute them, then assemble the results and plot, often with custom logic. The **dython** package automates much of this workflow: it inspects variable types, computes suitable association measures, returns a clean tabular result.

As dython was designed to be used for research, it puts an emphasis on the visual plots generated by its core methods, providing highly-readable and customizable visualizations of the output, treating those as a core component rather than a by-product.

In short, **dython** lowers the friction for inter-variable association analysis in mixed-type datasets and improves reproducibility of EDA workflows.

First version release date: February 2018.

Statement of Need

While there are many statistical and visualization libraries in Python (e.g. pandas (The pandas development team, 2025), `scipy` (Virtanen et al., 2020), `scikit-learn` (Pedregosa et al., 2011), seaborn (Waskom, 2021)), they treat continuous data, categorical data and the overall visualization separately. Users often resort to custom glue code to:

1. determine which columns are categorical vs numeric,
2. choose an appropriate association statistic (e.g. Pearson for numeric–numeric, correlation ratio for numeric–categorical, Cramér's V or Theil's U for categorical–categorical),
3. compute those pairwise,
4. assemble a matrix or graph,
5. annotate, visualize, and interpret the results.

This fragmentation results in boilerplate, inconsistency, or mistake-risk, especially in exploratory settings or pipelines.

dython addresses this gap by providing a unified, high-level API that:

- **infers variable types**
- **automatically selects appropriate measures**

- **returns structured and annotated output**
- **offers visualization** (heatmaps, annotation) integrated
- **offers model evaluation tools** (ROC, AUC, thresholding) for classification tasks

Therefore, **`dython`** helps data scientists, statisticians, and researchers spend less time writing glue code and more time focusing on insights.

Functionality

Below is a summary of existing methods of `dython`, per module.

`nominal`

Method	Description
associations	Computes associations between mixed-type features.
cluster_correlations	Applies clustering to reorder a correlation matrix.
compute_associations	Deprecated; replaced by associations(compute_only).
conditional_entropy	Computes conditional entropy of X given Y.
correlation_ratio	Computes correlation between categorical and numeric vars.
cramers_v	Computes Cramer's V between categorical variables.
identify_nominal_columns	Detects nominal (categorical) columns.
identify_numeric_columns	Detects numeric columns.
numerical_encoding	Encodes a mixed dataset into numeric format.
replot_last_associations	Re-plots the last association heatmap.
theils_u	Computes Theil's U (uncertainty coefficient).

`model_utils`

Method	Description
ks_abc	Computes KS statistic, ABC, and optional plot.
metric_graph	Plots ROC/PR curves for classifiers.
random_forest_feature_importance	Plots feature importance for Random Forest models.

`sampling`

Method	Description
boltzmann_sampling	Samples values under Boltzmann distribution.
weighted_sampling	Samples values using weighted probabilities.

`data_utils`

Method	Description
identify_columns_with_na	Returns dataset columns containing NA values.
identify_columns_by_type	Identifies columns of requested data types.

DRAFT

Method	Description
one_hot_encode	Converts a 1D array of integers into a one-hot matrix.
split_hist	Plots a histogram split by categories.

Code Examples

Associations

- `dython.nominal.associations(df, theil_u=False, plot=False, return_results=False, **kwargs)`
  Computes pairwise associations across all columns in a pandas DataFrame `df`. Internally, for each pair, it selects a measure appropriate to the variable types:
  - continuous–continuous → Pearson correlation (or Spearman, if configured)
  - continuous–categorical → correlation ratio
  - categorical–categorical → Cramér's V or Theil's U

  It outputs a pandas DataFrame (square matrix) of association values and optionally produces a heatmap (with annotations).

  Example usage:

  ```
  from dython.nominal import associations
  assoc_df = associations(my_df, theil_u=True, plot=True)
  ```

Model evaluation

- `dython.model_utils.metric_graph(y_true, y_pred, metric='roc', **kwargs)`
  This utility helps visualize classification performance. For a given true-label array y_true and predicted scores y_pred, it can plot ROC curves, compute AUC for each class (in multiclass settings), and show threshold recommendations.

  Example:

  ```
  from dython.model_utils import metric_graph
  metric_graph(y_true, y_pred_probs, metric='roc')
  ```

- `dython.model_utils.ks_abc(y_true, y_pred, **kwargs)` Perform the Kolmogorov–Smirnov test over the positive and negative distributions of a binary classifier, and compute the area between curves.

  Example:

  ```
  from dython.model_utils import ks_abc
  ks_abc(y_true, y_pred_probs)
  ```

Related work

Several libraries provide components somewhat overlapping `dython`'s functionality:

- `scipy.stats` (Virtanen et al., 2020), `statsmodels` (Seabold & Perktold, 2010) — full support for continuous correlations and tests, but limited categorical association tools
- `scikit-learn` (Pedregosa et al., 2011) — mutual information, label encoding, classification metrics, but lacks seamless cross-type association matrices

DRAFT

- `pingouin` (Vallat, 2018) — a statistical package including correlation, effect sizes, but does not integrate categorical–categorical measures like Theil's U or automatic visualization

Installation

You can install the released version via:

- **pip**: `pip install dython`
- **conda**: `conda install -c conda-forge dython`
- **Source**: `pip install git+https://github.com/shakedzy/dython.git`

Dependencies include standard scientific Python packages such as `numpy` (Harris et al., 2020), pandas (The pandas development team, 2025), `scipy` (Virtanen et al., 2020), `scikit-learn` (Pedregosa et al., 2011), `matplotlib` (Hunter, 2007), and `seaborn` (Waskom, 2021).

Usage Mention

Throughout its lifetime, `dython` has been used in many projects, including:

- Official implementation of TabDDPM (Kotelnikov et al., 2023) by Yandex Research
- `gretel-synthetics` by Gretal.ai
- `torchmetrics` by Lightning AI
- `ydata-quality` by YData

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Appendix: Code Examples

A minimal example using `associations`:

```
import pandas as pd
from sklearn import datasets
from dython.nominal import associations

# Load dataset
iris = datasets.load_iris()

# Convert int classes to strings to allow associations method
# to automatically recognize categorical columns
target = ["C{}".format(i) for i in iris.target]

# Prepare data
X = pd.DataFrame(data=iris.data, columns=iris.feature_names)
y = pd.DataFrame(data=target, columns=["target"])
df = pd.concat([X, y], axis=1)
```

DRAFT

```
# Plot features associations
associations(df)
```

This would output:

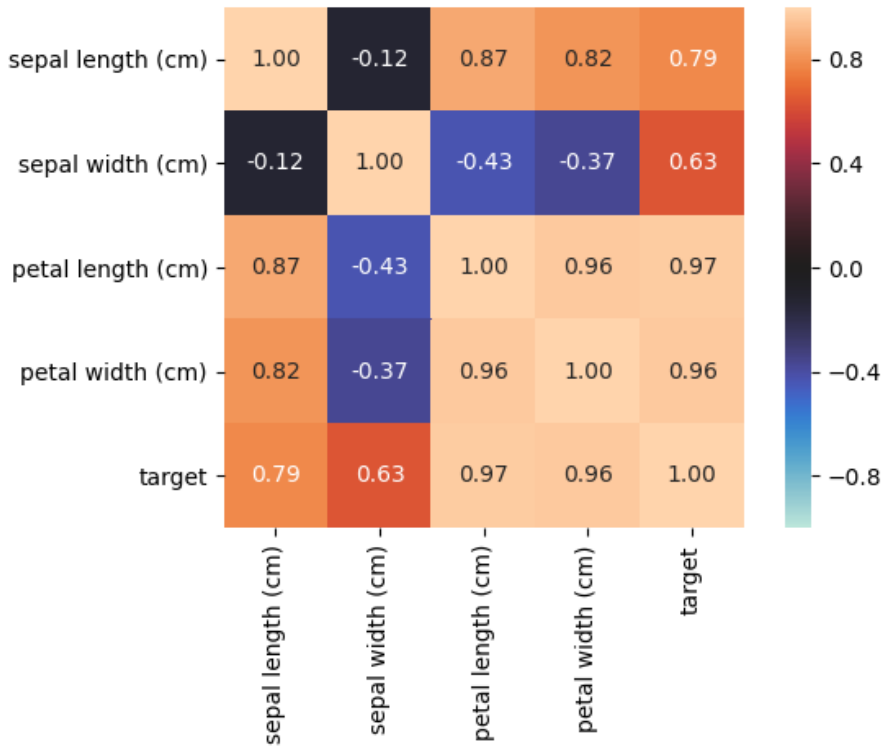

Figure 1: Example of an associations heatmap plotted over the Iris Dataset

A minimal example using `metric_graph`:

```
from sklearn import datasets, svm
from sklearn.preprocessing import label_binarize
from sklearn.model_selection import train_test_split
from sklearn.multiclass import OneVsRestClassifier
from dython.model_utils import metric_graph


# Load data
iris = datasets.load_iris()
X = iris.data
y = label_binarize(iris.target, classes=[0, 1, 2])

# Add noisy features
random_state = np.random.RandomState(4)
n_samples, n_features = X.shape
X = np.c_[X, random_state.randn(n_samples, 200 * n_features)]

# Train a model
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.5, random_state=0
)
```

```
classifier = OneVsRestClassifier(
    svm.SVC(kernel="linear", probability=True, random_state=0)
)

# Predict
y_score = classifier.fit(X_train, y_train).predict_proba(X_test)

# Plot ROC graphs
return metric_graph(
    y_test, y_score, "roc", class_names_list=iris.target_names
)
```

This would output:

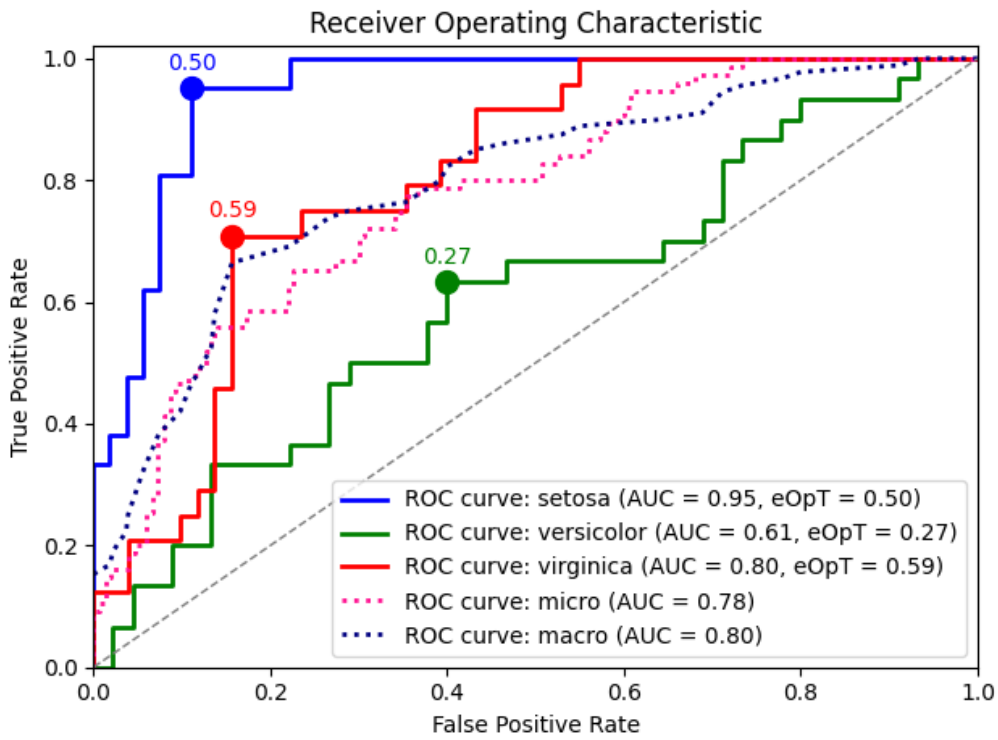

Figure 2: Example of a ROC graph plotted over the Iris Dataset

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